n = 83: Z[τ] closure + index lift, against the index-lift row, end-to-end cold

2026-10-19

n = 83: Z[τ] closure + index lift, against the index-lift row, end-to-end cold

Status (2026-10-19): frozen; Tier 1 run pending. The protocol and the schedule were committed before any Tier 1 timing. Every number below that is not from a Tier 1 run is either from the parents' archived Tier 1 runs (named each time) or from the cloud-agent VM, where it is a **local diagnostic**, not timing evidence.

Question

The n = 83 a = 1 baseline row is the **index-lift producer** (`koblitz-n83-indexlift-20261019`, PR #981, run 39290913632): the rank-probe producer with `KIC_INDEX_CANON=lift`. End-to-end cold is 165.2 s (shared rank stage 148.4 s + cold 16.8 s), `S` 6.37, gap to the tuned rho rT770 62.9.

The Z[τ]-multiplier closure (`koblitz-n83-closure-20261019`, branch `claude/n83-closure`, frozen commit 3761fd260, run 39305259093) measured 6.444 [6.099, 6.785] end to end. That comparison was against the **rank-probe** producer, not the current row, so it cannot replace the row: docs/ic/BASELINES.md asks for an improvement over the row on the same instances.

The two changes act on different stages. The closure cuts the rank stage's columns from 600 to 16. Index lift speeds up the root-index build, which every cold and rank process pays. They should stack. This run asks: does the closure, built on top of the index-lift producer, lower the **end-to-end cold** cost of the index-lift row on new targets and new rank seeds? Every relation must be checked in the group, every column log and scalar verified, the probe law kept, and the index-lift layout reproduced byte for byte. It is an iteration under AGENTS.md "Success criterion". A promoted result becomes the new n = 83 row.

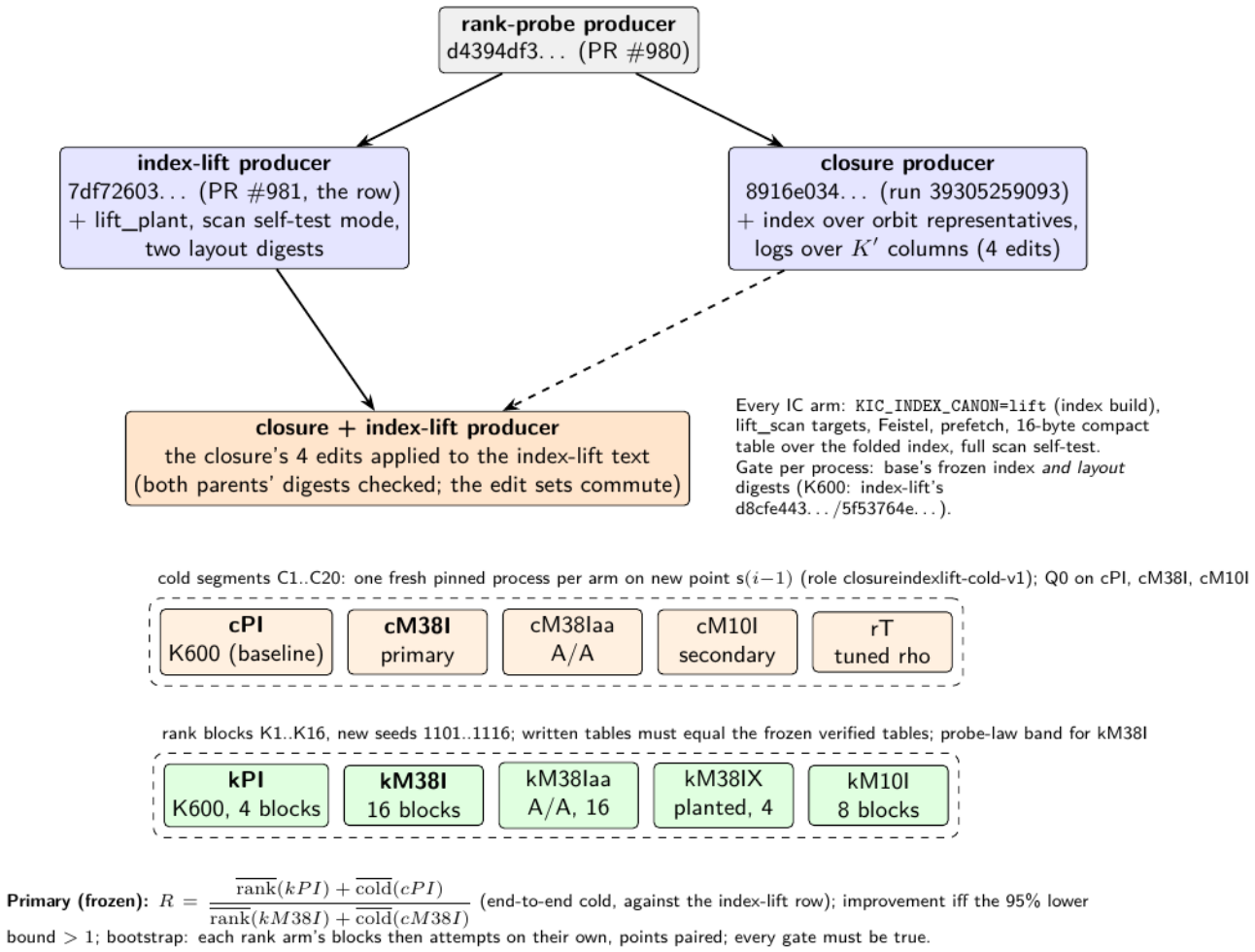


Figure 1: Design (from the frozen protocol; not a result). Source: figures/design.tex, vector: figures/design.pdf.

The producer: was a code change needed?

`KIC_INDEX_CANON=lift` is a runtime selector of the **rank-probe** producer (`INDEX_CANON_LIFT`, parsed from the environment). Both the closure producer and the index-lift producer derive from that producer, so the closure producer with `KIC_INDEX_CANON=lift` already builds the candidate's index. But the index-lift row's own producer adds two layout digests (`stored_states_blake3`, `table_slots_blake3`), `lift_plant` and `KIC_SCAN_SELFTEST`. The closure producer lacks the layout digests, so it cannot carry index-lift's layout gate, and it is not the row's binary.

So the producer is a **composition with no new code**. `make_online_source.py` applies the closure's four counted replacements (`closure_edits`, imported from `make_closure_source.py`, not re-typed) to the index-lift producer's text, plus one header comment. Both parents' derivations are verbatim files here, and each derivation's digest is checked on every build:

text	derivation	frozen sha256
rank-probe producer	`make_rankprobe_source.py` + five parents	`d4394df3...` (PR #980)
index-lift producer (the row)	`make_indexlift_source.py` (PR #981 verbatim)	`7df72603...`
closure producer	`make_closure_source.py` (3761fd260 verbatim)	`8916e034...`

text	derivation	frozen sha256
this producer	`make_online_source.py`	`1d328b2e...` (`protocol.json`)

The derivation also checks that the two edit sets commute, up to the header comments. Applying index-lift's nine edits to the closure text gives the same program. The diff from the index-lift producer is the closure's four replacements (59 diff lines). One binary, `ic_online_closure_indexlift_r770`, runs every IC arm, so the baseline and the candidate share the per-probe code.

Exactness checks (before freezing; local, cloud-agent VM)

Same layout as index-lift on B's header, and the closure's frozen identities on its bases (inputs/local_layout_test.json, `smoke/exactness_local.pycold`). Q0 ran with each base's verified table and both index canonicalizers, six processes:

base	logical-state blake3 / map digests / root entries	layout digests (stored states / table slots)	Q0 relation, scalar, probes
K600, selected and lift	B's `baf46d5a...`, 29,878,182	`d8cfe443...` / `5f53764e...` = **index-lift's frozen values**, both canonicalizers	B's archived, 1,987,852
M38, selected and lift	closure protocol's `09251580...`, 30,366,551	`f7a8d6b8...` / `6c08c6c2...`, one pair for both canonicalizers	closure's frozen, 1,612,533
M10, selected and lift	closure protocol's `ebf4f41e...`, 29,821,794	`c90b501b...` / `4973cfe5...`, one pair	closure's frozen, 2,130,165

The M38 and M10 layout pairs are frozen into `protocol.json` (`online.bases.*.layout_digest`). On every base, lift_scan builds the same slot array as the selected canonicalizer. This is index-lift's argument, which now holds on the closure bases as well.

Archived Tier 1 rank processes reproduced exactly with `KIC_INDEX_CANON=lift` (smoke/exactness_rank_local.json):

- **kPI of the index-lift run** (block K1, seed 401, point s000 of `indexlift-cold-v1`, B's header). The relation digest, all 600 per-attempt probe counts, root calls, point-addition and frame counters, index and layout digests, target relation, scalar and probes are equal. The written table equals B's.
- **kM38 of the closure run** (block K1, seed 801, point s000 of `closure-cold-v1`, M38 header). This archived process used the selected index build. The relation digest, all 16 per-attempt probe counts, root calls, counters, index digests, target relation, scalar and probes are equal. The written table equals the frozen verified M38 table.

So the derived producer reproduces index-lift on B's header (cold and rank) and the closure on M38 (cold and rank), field for field.

Tests (smoke/cargo-test-online.txt): all 21 producer tests pass, including the closure's `closure_multiplier_is_its_label` and index-lift's two layout tests, plus the 4 tuned-rho tests. The 11 verifier unit tests (`test_verify_receipt.py`) pass. The workflow runs both before any timing.

Frozen Tier 1 design

One GitHub `ubuntu-24.04` runner, one pinned CPU, Tier 1 counters, one fresh process per run. Every IC arm runs `ic_online_closure_indexlift_r770` with:

- `KIC_INDEX_CANON=lift`;
- the full scan self-test;

- lift_scan target keys, the Feistel order and the prefetch pipeline;
- the 16-byte compact table over the folded index;
- the frame rank loop.

Arm	Kind	Base	Role
`cPI`	cold	K600 + B's table	**baseline** cold (the index-lift row's cPI)
`kPI`	rank	K600	**baseline** rank stage (the row's kPI; 600 attempts), blocks K1, K5, K9, K13
`cM38I`	cold	M38 + its verified table	**primary candidate** cold
`kM38I`	rank	M38	**primary candidate** rank stage (16 attempts), all 16 blocks
`cM38Iaa`, `kM38Iaa`	cold, rank	M38	A/A of the primary
`kM38IX`	rank	M38	planted positive control (`frame_plant`, about 1.8x per attempt in run 39305259093), blocks K1, K5, K9, K13
`cM10I`, `kM10I`	cold, rank	M10	pre-registered secondary (60 attempts), the 8 odd blocks; never decides
`rT`	rho	—	tuned rho rT770 (Fold/64/DP6), seed 1010000 + point seed

- **Cold panel.** 20 new public points (`inputs/cold_targets.json`, role `closureindexlift-cold-v1`, `make_targets.py`). They are disjoint from:
  - every earlier n = 83 panel of the law (Q0, B's panel, scan-order, combined, probe-pipeline, state-fold, table-filter, stacked, rank-cut, rank-probe, index-lift, the 64 n131 `eval` points);
  - the panels of the lanes n83-rotkey, n83-k899, n83-onlineframe, n83-closure, n83-knownlog and n83-blrp (`rotkey-cold-v1`, `k899-cold-v1`, `onlineframe-v1`, `closure-cold-v1`, `knownlog-cold-v1`, `blrankprobe-cold-v1`).

  Every panel was regenerated and checked equal to the file committed in its worktree. A role-agnostic scan of 211 input files in those worktrees (13,731 points) found no shared point. No x-coordinate of the panel occurs in origin/main's `experiments/` or `research/`. No scalar is known.
- **Rank blocks.** K1–K16 use the new seeds 1101–1116. Already used on n = 83: 7, 101–102, 201–204, 301–309, 401–404, 501–504, 701, 801–816, 901–904.
- **Order.** O1 (Q0 on cPI, cM38I and cM10I), then C1 K1 … C16 K16 C17 … C20. The order within each segment is shuffled with new seeds. 151 runs; time guard 300 minutes.

Primary (pre-registered).

- R = E2E(kPI, cPI) / E2E(kM38I, cM38I), where E2E(k, c) = the mean rank stage of k over its blocks plus the mean cold time (setup + one-target online) of c over the 20 points.
- Bootstrap: 10,000 draws, seed 20261219. Each rank arm's blocks are resampled, then its attempts within each drawn block, independently for the two arms. The points are resampled, paired across cPI and cM38I.
- **Improvement** iff the 95% lower bound > 1; regression iff the upper bound < 1; otherwise not separated.
- **Invalid** if any gate is false *or unknown*, any control fails, or any scheduled row is missing.

Controls.

- A/A: E2E kM38I+cM38I / kM38Iaa+cM38Iaa and cold cM38I / cM38Iaa must both be inside [0.97, 1.03].
- Planted timing control: the per-attempt ratio kM38IX / kM38I must have its lower bound above 1.
- Planted exactness control: the three closure corruptions (a label, a duplicated orbit, a log) must be rejected by `verify_closure.py` on both closure bases before timing.

Gates. These are the closure experiment's gates plus index-lift's layout gate:

1. Before timing (`check_log_table.py`):
   - B's table re-verified in the pinned oracle;
   - each closure header regenerated to its frozen SHA-256 and passed by `verify_closure.py`, with the planted corruptions rejected;
   - each closure table checked: [L_c]G = R_c for every column and [w·L_c]G = α(R_c) for every orbit;
   - the closure logs equal B's archived logs of the same R_c.
2. Every IC process:
   - `KIC_INDEX_CANON=lift` recorded, 0 roots checked against the selected canonicalizer, and the full scan self-test;
   - its base's frozen base hash, columns, orbits, states, slots, root entries, logical-state and table-map digests;
   - **its base's frozen layout digests.** For the baseline arms these are index-lift's `d8cfe443...` / `5f53764e...`.
3. Per point: one scalar across every IC process (all three bases) and rT, replayed in the oracle. One relation and one probe count per point and base. One index and layout digest set per base, equal to the frozen one.
4. Q0: cPI gives B's archived identity (1,987,852 probes). cM38I and cM10I give the closure protocol's frozen Q0 identities.
5. Rank arms:
   - full rank, 0 failures, one frame per attempt;
   - the written table equal to the base's frozen verified table, log for log;
   - per block, identical relation digests, probe vectors, root calls and counters across kM38I, kM38Iaa and kM38IX.
6. Probe law: kM38I's pooled probes per relation over r / (166 E) must lie in [0.80, 1.20] over 256 attempts. kM10I's band is reported only.
7. Receipt integrity, fail closed (index-lift's hardened verifier on main):
   - every raw record decodes;
   - its SHA-256 equals the wrapper's `raw_result_sha256`;
   - its arm, kind, panel, point and seed equal the schedule's;
   - an unknown gate is a failure.

Power

The power check is power_check.py, with results in analysis/power.json. Its prior, `inputs/power_prior.json`, is built by `make_power_prior.py` from the parents' archived Tier 1 runs:

- **Closure run 39305259093:** the kP blocks (600 attempts), the kM38 / kM38aa blocks (16 attempts; their ratios are the process factors), the kM10 blocks, and the cP / cM38 / cM10 cold times.
- **Index-lift run 39290913632:** the cold factor L = mean cPI / mean cP = 0.7476, applied to every cold arm here, since all of them build with lift_scan.

There were 200 simulations per cell, each with 300 bootstrap draws.

end-to-end effect	M38 (16 blocks)	M10 (8 blocks)
1.00 (no gain; type I)	0.005	0.02
1.10	0.15	0.275
1.50 (minimal effect of interest)	**0.995**	**1.00**
2.00	1.00	1.00
4.00	1.00	1.00
modelled (both changes as measured)	1.00 (median R 7.98)	1.00 (median R 5.13)

Expected effect. At the parents' archived means the model gives (108.0 + 12.0) / (2.75 + 12.3) s = **about 8.0** for M38 and 5.1 for M10. This is an estimate, not a claim. It is higher than the closure's 6.4 because index lift shrinks the cold path, which is most of the candidate's remaining cost, and it uses the closure run's faster runner. The

probe-law gate passes at the true law with probability 0.9955. It rejects a law 1.5x worse with probability 0.9965 (256 attempts; archived kM38 ratio 0.962).

Runtime estimate. About 1.1 hours: build and tests 15 min, cold segment about 55 s, rank block about 190 s with kPI and about 45 s without.

Local runner smoke (correctness only)

smoke/README.md and smoke/correctness_smoke.json: segments O1, C1 and K1 through `run_schedule.py` and `arm.py`, then `verify_receipt.py --correctness-only`.

- `check_log_table.py` passes for K600, M38 and M10, and the planted corruptions are caught.
- All 13 rows are valid, covering every arm.
- Every correctness check passes on them:
  - one scalar per point across all three bases and rT, replayed in the oracle;
  - Q0 equal to the archived and frozen identities on all three bases;
  - one index and layout digest set per base, equal to the frozen one;
  - identical rank digests and probe vectors across kM38I, kM38Iaa and kM38IX;
  - every rank table equal to its frozen verified table.
- The one exception is the probe-law band on this single block (kM38I 1.566 over 16 attempts). The frozen gate pools 256 attempts. The closure's per-block ratios ranged from 0.58 to 1.35, and its run pooled 0.963.
- The protocol was frozen before this smoke and was not changed after it. The smoke used the frozen block K1. Its probe counts are deterministic and will recur in Tier 1.

Result

Pending: the Tier 1 run of `.github/workflows/n83-closure-indexlift.yml` (dispatch only), at the frozen commit. `verify_receipt.py` applies the frozen rule to its artifact, and `make_figures.py` renders the result figure from `result.json`.

Not included

- Factor-base construction is charged to no arm, the convention of every n = 83 row. A closure header holds B's first K' orbits plus (M − 1)·K' images α(R_c), each costing two small scalar multiplications.
- index-lift's secondary cPIT (sampled scan self-test) and its planted `lift_plant` are not arms. Here every arm runs the full self-test, and the layout gate checks the index-lift change on every process.

Files

File	Role
`protocol.json`, `make_protocol.py`, `schedule.json`, `make_schedule.py`	frozen protocol, its builder, frozen run order
`make_online_source.py`	the composition: closure edits on the index-lift text; checks all three parent digests and that the edits commute
`make_indexlift_source.py`, `make_closure_source.py`, `make_rankprobe_source.py` + five parents, `online/`, `lift2/`	the parents' derivations, verbatim; the crate; the two-limb canonicalizer
`make_closure_base.py`, `verify_closure.py`, `inputs/factor_logs_n83_closure_*.json`, `inputs/local_decisive_test.json`, `inputs/parent_closure_protocol.json`	the closure experiment's header builder, verifier, verified tables, local record and frozen protocol (commit 3761fd260), verbatim
`inputs/local_layout_test.json`, `smoke/exactness_local.py`, `smoke/exactness_rank_local.json`	the exactness checks above

File	Role
restore.py, check_log_table.py	restore and digest-check crypto 770f6a69, the closure headers and every local file; verify every table and closure before timing
make_targets.py, inputs/cold_targets.json	the new cold panel
make_power_prior.py, inputs/power_prior.json, power_check.py, analysis/power.json	power prior (runs 39305259093 and 39290913632), simulation, result
arm.py, run_schedule.py	one arm per pinned process; the schedule runner
verify_receipt.py, test_verify_receipt.py	gates, statistics, fail-closed decision rule; its tests
figures/design.*, make_figures.py	design diagram; result figure after the run
README.pdf	this report: pandocREADME.md-oREADME.pdf--pdf-engine=pdflatex-Hreport_header.tex--lua-filterreport_filter.lua-Vgeometry:margin=2cm-Vfontsize=10pt-Vcolorlinks=true